

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Bachelor of Technology (Electronics and Computer Engineering) SEMESTER - 5 Summer 2025 (Supply.)

Course :Bachelor of Technology (Electronics and Computer Engineering) Branch : Engineering and Technology

Semester : SEMESTER - 5

Subject Code & Name: BTECOE504C - PROGRAMMING IN JAVA

Time : 3 Hours]

[Total Marks : 60

Instructions to the Students:

1. Each question carries 12 marks.
 2. Question No. 1 will be compulsory and include objective-type questions.
 3. Candidates are required to attempt any four questions from Question No. 2 to Question No.6
 4. Use of non-programmable scientific calculators is allowed.
 5. Assume suitable data wherever necessary and mention it clearly.
-

Q1. Objective type questions. (Compulsory Question)

12

- 1 Java is a:
 - a) Compiled language only
 - b) Interpreted language only
 - c) Both compiled and interpreted
 - d) Assembled language
- 2 Which of the following is used to create an object in Java?
 - a) create
 - b) new
 - c) alloc
 - d) construct
- 3 What is the purpose of the 'this' keyword in Java?
 - a) Refers to the superclass
 - b) Refers to the current class
 - c) Refers to the current object
 - d) Refers to the child class
- 4 Which method is called automatically when an object is created?
 - a) finalize()
 - b) main()
 - c) init()
 - d) Constructor
- 5 What is the use of the 'implements' keyword?
 - a) To inherit from a class
 - b) To define a package
 - c) To implement an interface
 - d) To import a class
- 6 Can a class implement multiple interfaces in Java?
 - a) No
 - b) Only if interfaces are final
 - c) Yes
 - d) Only one interface at a time

7 Which interface must be implemented to create a thread in Java?
a) Runnable
b) Threadable
c) Callable
d) EventListener

8 Which keyword is used to handle exceptions in Java?
a) try-catch
b) exception
c) catch-throw
d) error-handle

9 Which package is primarily used for graphics in Java?
a) java.awt
b) java.io
c) java.util
d) java.graphics

10 Which method is used to load a JDBC driver class in Java?
a) DriverManager.connect()
b) Class.forName()
c) loadDriver()
d) Driver.load()

11 Which one of the following is not an access modifier?
a) Protected
b) Void
c) Public
d) Private

12 What is the output of the following code?

```
class A {  
    int x = 10;  
}  
  
public class Test {  
    public static void main(String[] args) {  
        A obj = new A();  
        System.out.println(obj.x);  
    }  
}
```


a) x
b) 0
c) 10
d) Compile error

Q2. Solve the following.

A) Differentiate between C++ and Java.

6

B) What is variable scope in Java? Discuss local, instance, and static variables with suitable examples.

6

Q3. Solve the following.

A) Explain the concept of 'this' keyword in Java. How is it used to refer to current class members and constructor chaining?

6

B) Explain the use of the `final` keyword in Java with respect to variables, methods, and classes. What restrictions does it impose in each case?

6

Q4. Solve Any Two of the following.

A) State the difference between inheritance and interface in java.

6

B) Explain the concept of method overriding in Java. Illustrate with an example.

6

- C) What is inheritance in Java? List the types of inheritance supported in Java and explain multilevel inheritance with a suitable example. 6

Q5. Solve Any Two of the following.

- A) What is multithreading in Java? Explain the lifecycle of a thread with a suitable diagram. 6
- B) What is exception handling in Java? Explain the use of try, catch, and finally, keywords with an example. 6
- C) Differentiate between checked and unchecked exceptions in Java. 6

Q6. Solve Any Two of the following.

- A) What is JDBC in Java? Explain the steps to connect a Java application to a database using JDBC with example. 6
- B) What is the Abstract Window Toolkit (AWT) in Java? Explain the key features of the AWT package. 6
- C) Explain the Graphics class in Java. How is it used to draw shapes and text in GUI applications? 6

*** End ***